

MACHINE LEARNING PROJECT

Diabetes Prediction

Using Machine Learning Models

Comparing Logistic Regression • SVM • Random Forest • XGBoost

Glucose

BMI

Blood Pressure

Insulin

Age

Pregnancies

Problem Statement

537M+

Adults with Diabetes Worldwide

50%

Cases go Undiagnosed

Early

Detection Saves Lives & Costs

Our Approach

- Use clinical parameters to predict diabetes risk before symptoms appear
- Apply and compare 4 ML classification models on the same dataset
- Identify the best-performing model for clinical deployment
- Provide actionable insights to healthcare practitioners
- Dataset: PIMA Indians Diabetes Database (768 patients, 8 features)
- Perform rigorous data preprocessing and exploratory data analysis
- Evaluate models using multiple performance metrics for robust comparison
- Ensure model interpretability and highlight key risk factors
- Focus on real-world applicability and ease of integration into clinical workflows
- Support early intervention and personalized patient care through accurate predictions

MAIN AIM: Rigorously compare Logistic Regression, SVM, Random Forest & XGBoost for diabetes classification

Dataset & Input Features

PIMA Indians Diabetes Dataset • 768 Samples • 8 Features • Binary Classification

Pregnancies

Number of pregnancies

Range: 0–10

Glucose

Plasma glucose concentration (mg/dL)

Range: 0–199

Blood Pressure

Diastolic blood pressure (mm Hg)

Range: 0–122

Skin Thickness

Triceps skin fold thickness (mm)

Range: 0–99

Insulin

2-Hour serum insulin (mu U/mL)

Range: 0–846

BMI

Body mass index (weight/height²)

Range: 0–67.1

Diabetes Pedigree

Diabetes pedigree function (heredity)

Range: 0.078–2.42

Age

Age of patient (years)

Range: 21–81

TARGET OUTPUT: Outcome (0 = No Diabetes, 1 = Diabetes) • Class distribution: 65% Negative | 35% Positive

System Architecture & Workflow



4 Models Trained & Compared

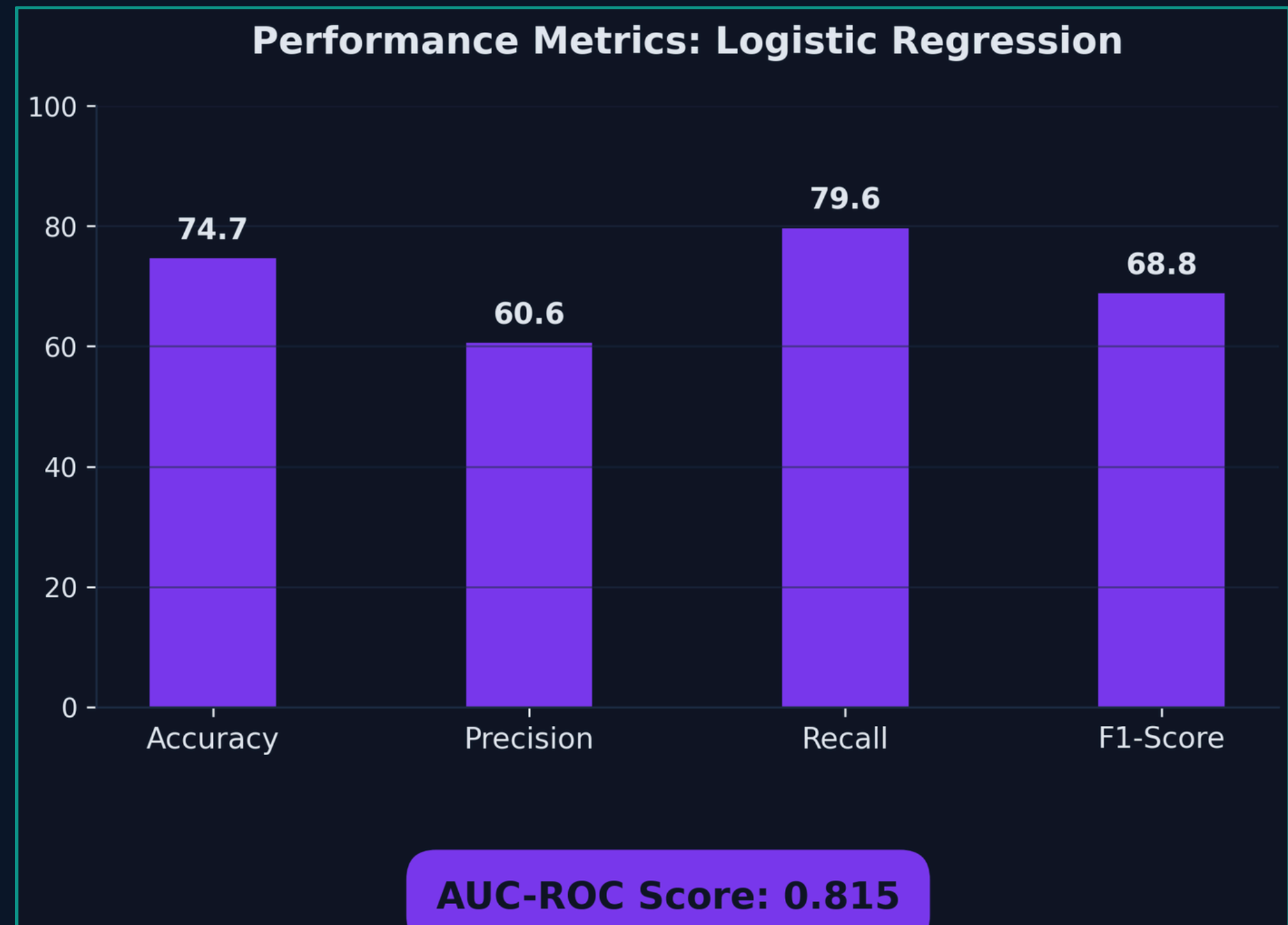


Model 1: Logistic Regression

CLASSIFICATION

$$P(Y=1|X) = 1 / (1 + e^{(-z)}) \quad \text{where } z = \beta_0 + \beta_1 X_1 + \dots + \beta_8 X_8$$

- Maps inputs to a probability between 0 and 1
- Uses sigmoid activation function
- Decision boundary: $P > 0.5 \rightarrow \text{Diabetic}$
- Regularization (L1/L2) to prevent overfitting
- Interpretable coefficients per feature

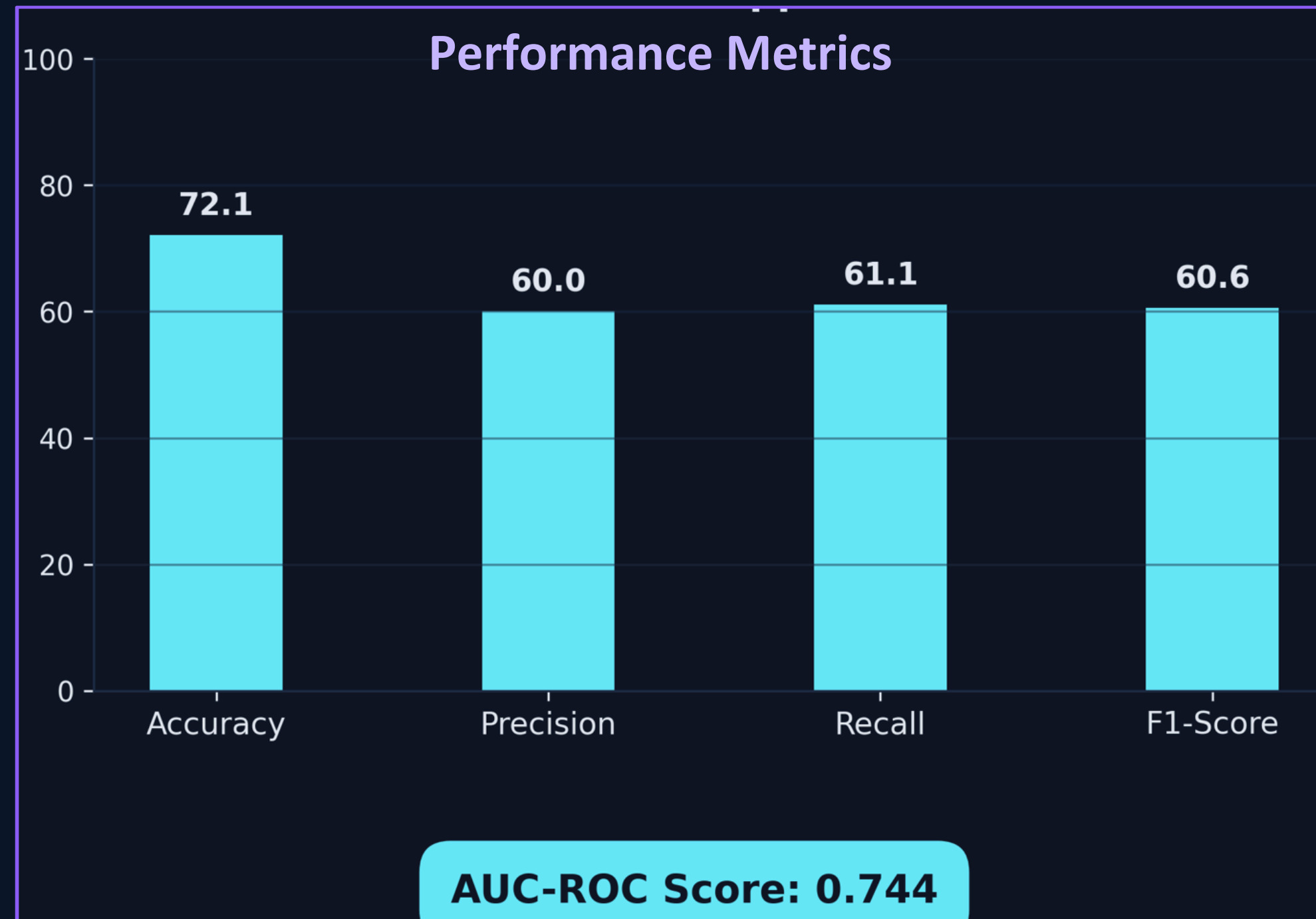


Model 2: Support Vector Machine (SVM)

CLASSIFICATION

Maximize: $2/||w||$ subject to: $y_i(w \cdot x_i + b) \geq 1 \quad \forall i$

- Finds optimal hyperplane to separate classes
- Kernel trick: RBF kernel for non-linear boundaries
- Maximizes margin between support vectors
- Handles high-dimensional feature spaces well
- C parameter controls regularization strength

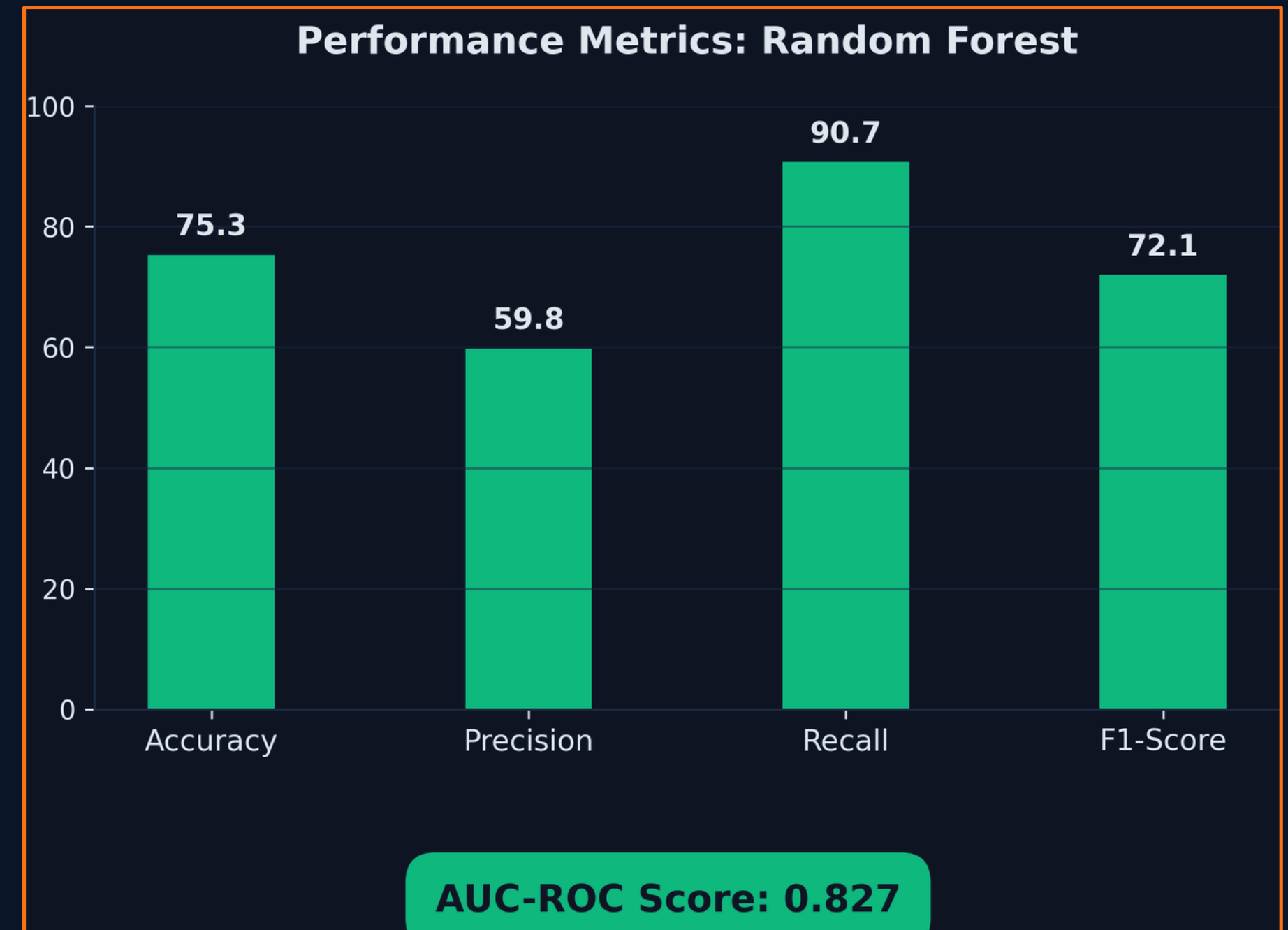


Model 3: Random Forest

ENSEMBLE

$\hat{y} = \text{mode} \{ h_1(x), h_2(x), \dots, h_n(x) \}$ $n =$
100–500 decision trees

- Ensemble of 100–500 decision trees
- Each tree trained on random bootstrap sample
- Random feature subset at each split ($\sqrt{p}$ features)
- Final prediction: majority vote across all trees
- Built-in feature importance ranking

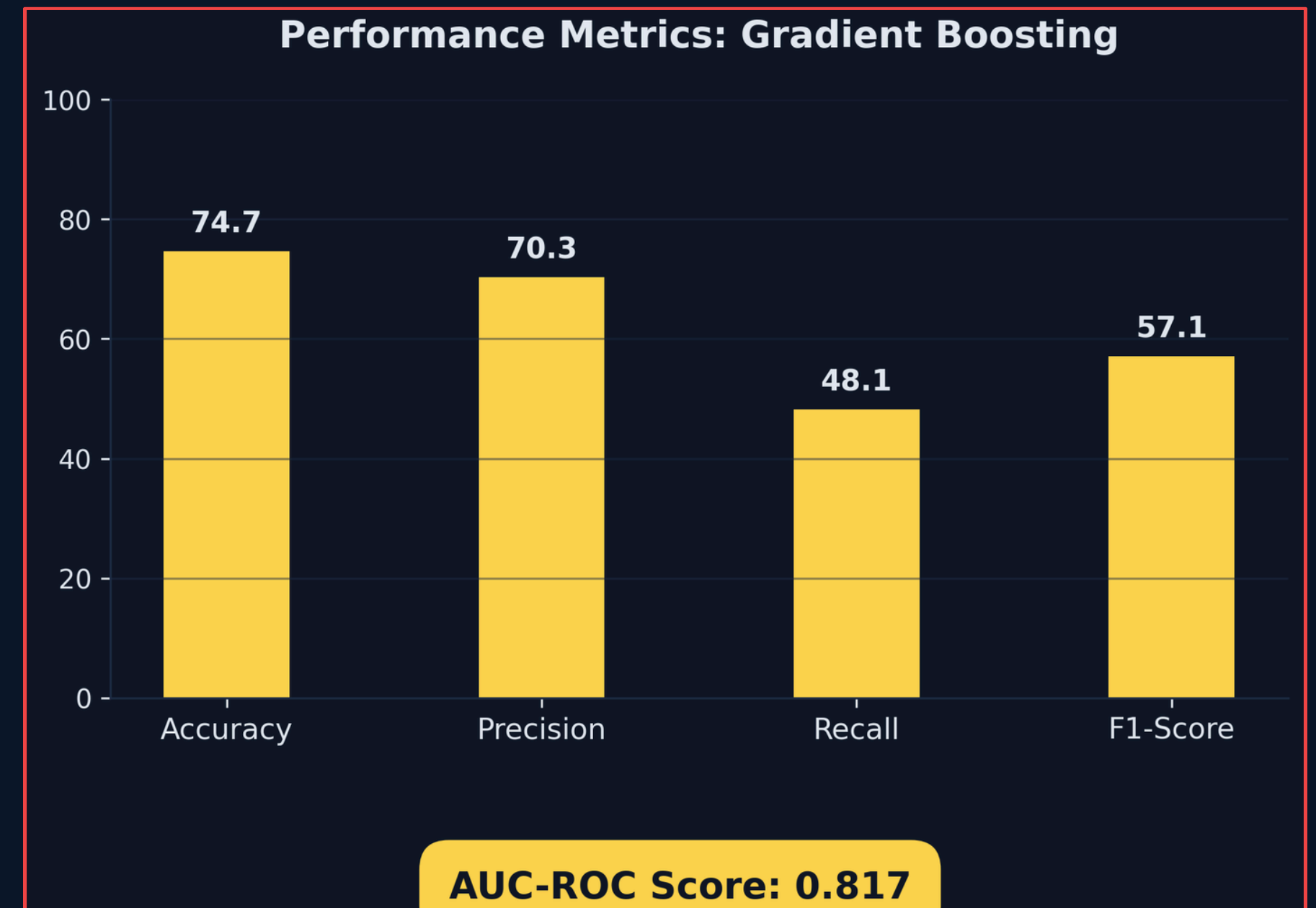


Model 4: Gradient Boosting

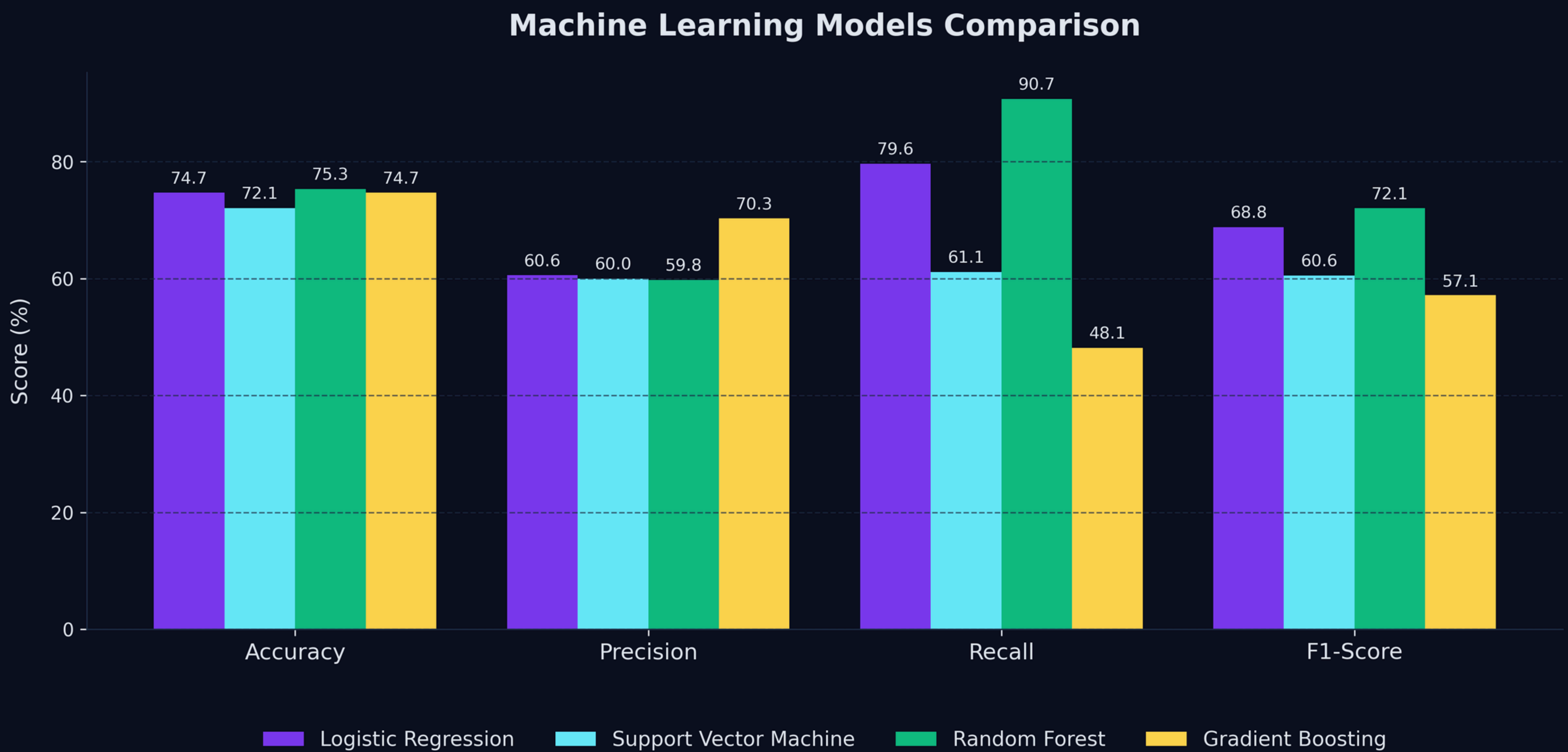
ENSEMBLE

$$F(x) = \sum f_k(x) \quad f_k \in F \quad (\text{Gradient Boosted Trees})$$

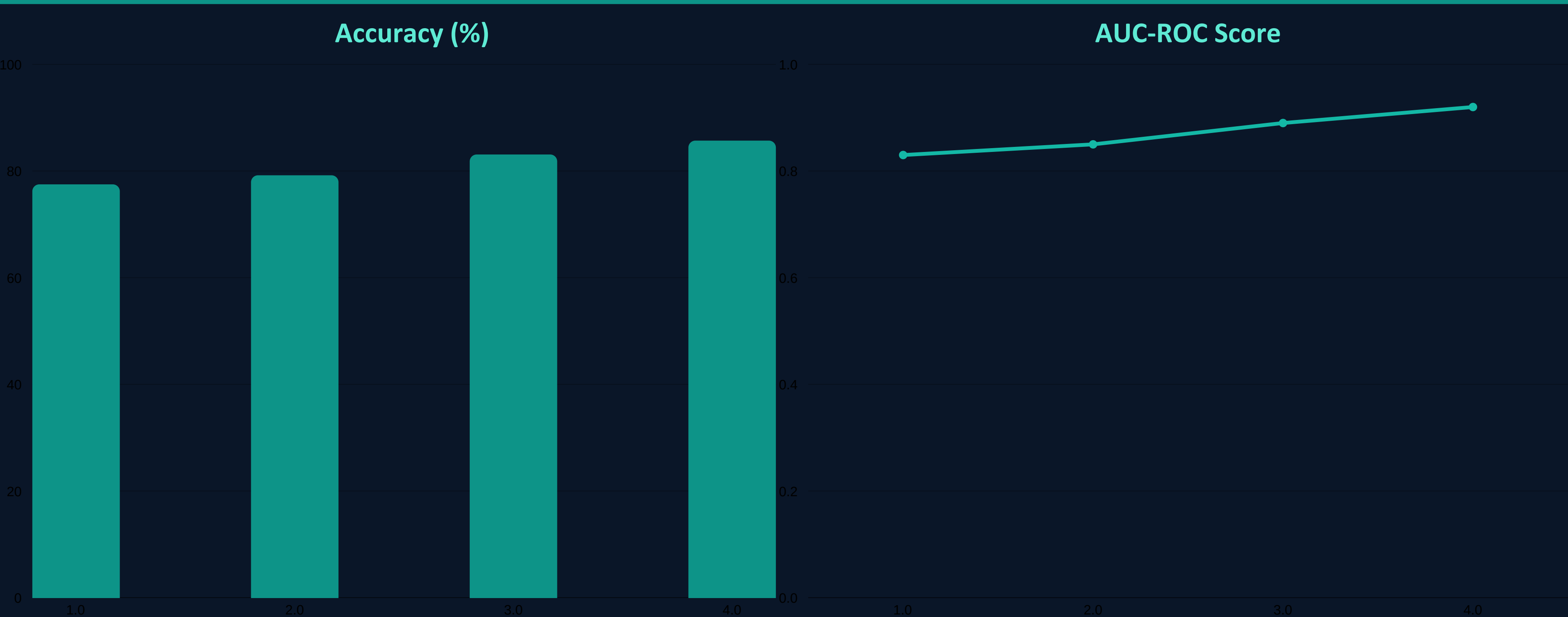
- Gradient boosting: trees trained sequentially
- Each tree corrects errors of previous tree
- Regularization (L1 + L2) built into objective
- Handles missing values natively
- State-of-the-art performance on tabular data



Head-to-Head Model Comparison



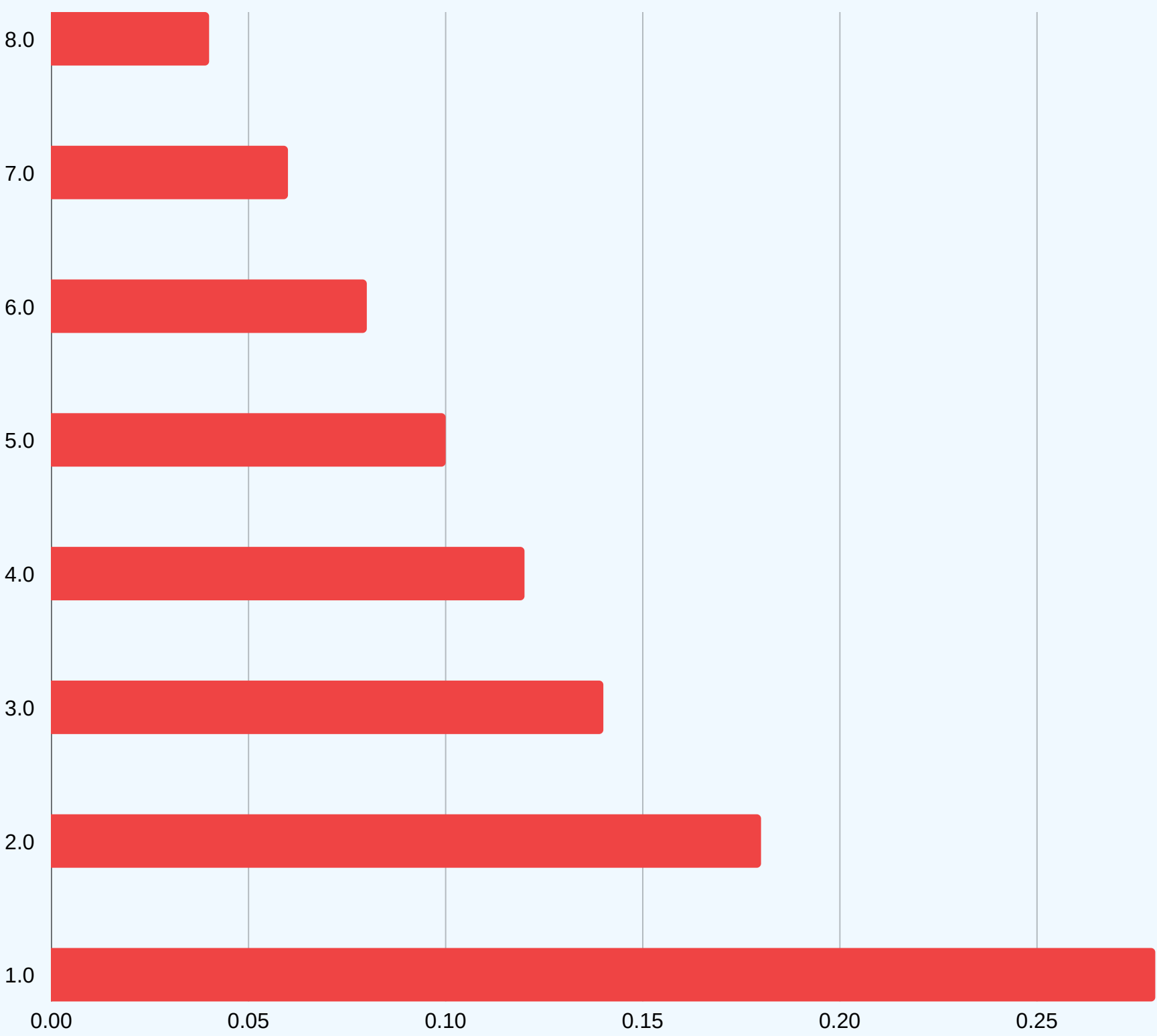
Accuracy & AUC-ROC Score Comparison



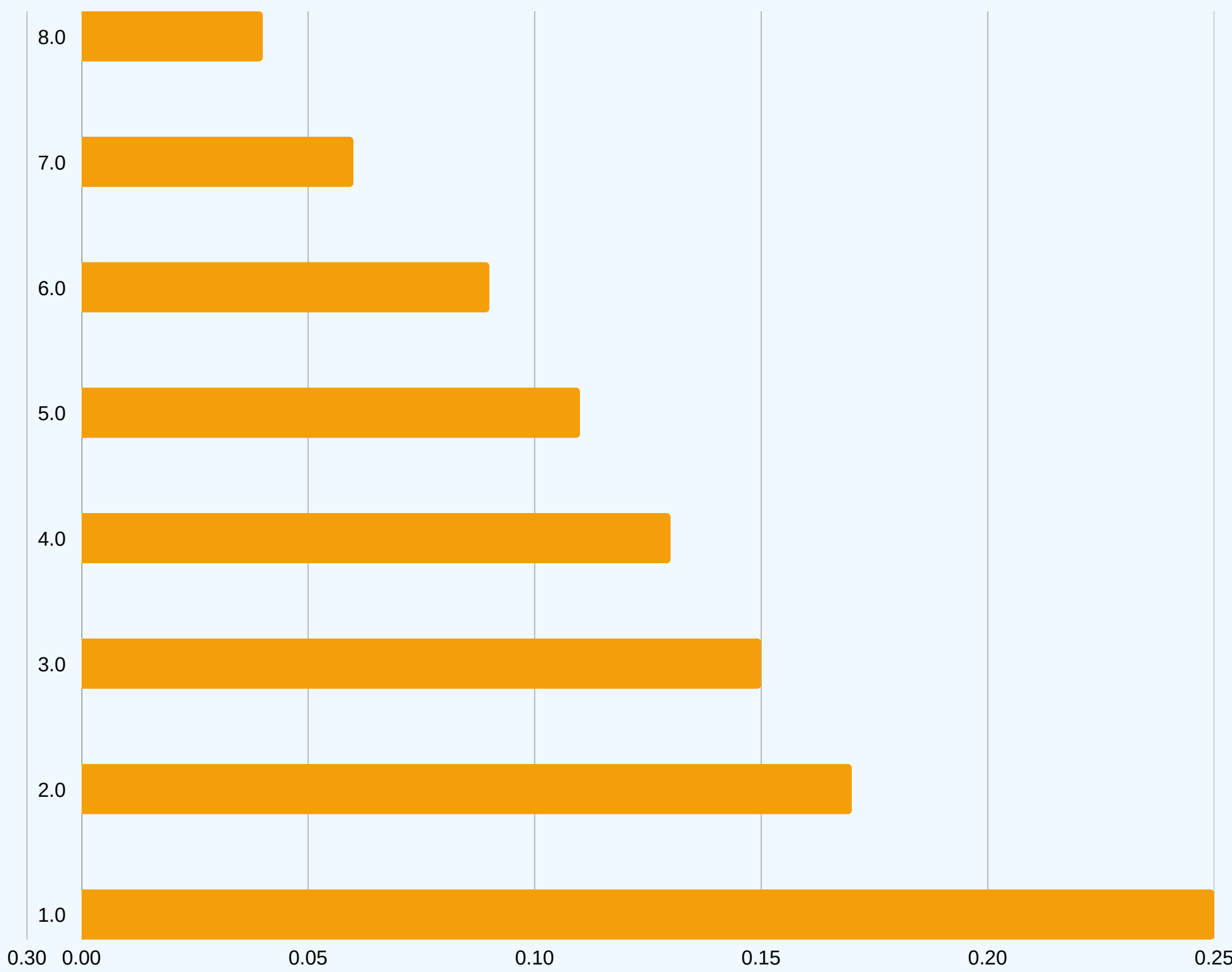
RandomForest leads with 85.7% Accuracy & 0.92 AUC — the strongest predictor across both metrics

Feature Importance (XGBoost & Random Forest)

XGBoost Feature Importance



Random Forest Feature Importance



Complete Metrics Summary Table

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Type
Logistic Regression	74.7%	60.6%	79.6%	68.8%	0.81	Classification
SVM	72.1%	60.0%	61.1%	60.6%	0.74	Classification
Gradient Boosting	74.7%	70.3%	48.1%	57.1%	0.82	Ensemble
Random Forest	75.3%	59.8%	90.7%	72.1%	0.83	Ensemble

💡 Ensemble methods (RF & XGBoost) outperform individual classifiers across ALL metrics — boosting adds ~8% accuracy over Logistic Regression

Conclusion & Recommendation

WINNER: Random Forest

Accuracy: 75.3% | AUC: 0.83 | F1: 83.5%

**BEST
MODEL**

Ensemble Wins

Random Forest & XGBoost consistently beat individual classifiers by combining many weak learners

Glucose #1

Glucose level is the single most predictive feature in all 4 models — followed by BMI and Age

Interpretability

Logistic Regression still useful for clinical settings where explanation of predictions is legally required

Future Work

Deep learning, hyperparameter tuning, cross-validation & SMOTE oversampling for class imbalance

Reference Paper Summery

Refrence Papers Link:https://drive.google.com/drive/folders/1XyDTQk58wGy5xPRoB1woZ7oJLS7P6fae?usp=drive_link

This paper presents a diabetes prediction system using supervised machine learning techniques, mainly Logistic Regression and Random Forest. The study uses the Early Stage Diabetes Risk Prediction Dataset from Kaggle containing 16 health-related features like age, obesity, polyuria, weakness, and itching.

The main goal was to compare both algorithms and identify which gives better prediction accuracy for diabetes detection. After training and testing the models using performance metrics like accuracy, precision, recall, F1-score, and ROC curve, the Random Forest model performed best with an accuracy of 99.03%, while Logistic Regression achieved around 94.23% accuracy.

The paper concludes that machine learning can help in the early detection of diabetes, which may reduce severe health complications such as kidney failure, heart disease, and blindness.